Topology of Neural-Network Layers for Out-of-Distribution Detection

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At a glance

Task: find a model used in the six papers of the literature report, implement all six approaches on it, use them to detect out-of-distribution (OOD) inputs, and compare how well each works.

Model: the Bayesian MLP 784→64→16→10 from the pTU paper [3], trained on MNIST (97.2 % test accuracy, close to the paper's 97.4 %).

Test: can each method tell MNIST digits apart from 7 kinds of unfamiliar images? Every result is averaged over 3 independently trained models.

Main result: 5 of the 6 topological approaches beat the standard confidence baseline. The best one, [5], reaches 96.5 / 100 AUROC versus 74.1 for confidence. Neural Persistence [1] cannot detect OOD inputs at all.

1. The shared model

- **Why this model:** papers [1], [2] and [3] all study small fully connected networks (MLPs) on MNIST, and [3] gives the exact architecture, prior and training rule, so the setup can be reproduced.
- **What it is:** 784 inputs (28×28 pixels) → 64 → 16 → 10 classes, ReLU, trained as a Bayesian network (Bayes-by-Backprop), so each weight has a distribution, not a single value.
- **How it is used:** methods [1], [2], [4], [5], [6] use the average weights; [3] (pTU) uses 30 random weight samples, as in the paper.
- **One change from the paper:** the KL term of the loss is scaled by 0.1. With the full term, accuracy stalls at 94.5 %, and [3] does not say how it weighted this term.

2. The six approaches

All six turn a layer of the network into a shape (a graph or a cloud of points), track how that shape connects up as a threshold changes (persistent homology), and summarize it with numbers. An input whose numbers look unusual compared with training data is flagged as OOD.

Paper	Main idea in one sentence	OOD score used here
[1] Neural Persistence Rieck et al. 2019	How strongly a few weights stand out in each layer of the trained network (uses weights only, no input).	Same number computed on each input's activations (my adaptation, since the original ignores the input).
[2] Topological Uncertainty Lacombe et al. 2021	Each input activates the network in a pattern; compare that pattern's topology with typical training inputs.	Distance to the average training pattern of the predicted class.
[3] pTU Yeh & Yang 2025	Run the input through 30 sampled versions of a Bayesian network; if their topologies disagree, the input is uncertain.	Spread (variance) of the 30 topologies; plus the paper's statistical test on groups of images.
[4] Persistent Laplacian Wolf et al. 2025	Eigenvalues of a matrix built from the shape: zero eigenvalues = topology, nonzero ones = extra geometry.	Number of zero eigenvalues (β_0) and the nonzero eigenvalues at 3 thresholds.
[5] Data manifold Magai & Ayzenberg 2022	Treat all images at one layer as a point cloud and measure its loops, clusters and dimension.	Groups of images: cloud topology + dimension. Single image (my adaptation): how much it changes its 20 nearest training neighbours' topology.
[6] CNN activations Purvine et al. 2023	Compare the topology of different layers' point clouds with a distance between diagrams.	Groups of images: distance from the group's diagrams to MNIST's diagrams.

3. How the methods were tested

- **Normal data:** 2000 MNIST test digits. **OOD data** (2000 each, all shown as 28×28 grayscale): Fashion-MNIST (clothes), KMNIST (Japanese characters), CIFAR-10 (photos), SVHN (house numbers), Gaussian noise, uniform noise, and MNIST-flip (digits with 100 pixels inverted).
- **Score:** AUROC = how well a method ranks OOD images above normal ones. **50 = random guessing, 100 = perfect.** FPR95 (lower is better) = % of OOD images missed when 95 % of normal images are accepted.
- **Two settings: one image at a time** (the usual OOD task) and **groups of images** (is this whole batch from a different source?), which is how papers [3], [5], [6] use topology.
- **Baselines:** softmax confidence (MSP, the baseline in [2]), Bayesian mutual information, and distance to the nearest training images (k-NN).

4. Results

4a. One image at a time — average over the 7 OOD sets, mean ± std over 3 trained models, ranked from best to worst. Grey rows are baselines.

#	Method	AUROC ↑	FPR95 ↓	Verdict
1	[5] Data manifold, single-image version	**96.5 ± 0.4**	**16.9 ± 2.3**	**Best** — but it mainly measures distance to the training data
2	k-NN distance (baseline)	88.4 ± 0.9	46.4 ± 1.9	**Strong baseline**
3	[2] TU with layers rescaled (my fix)	87.4 ± 0.6	60.1 ± 1.6	**Good** — +8 points over the paper's version
4	[4] Persistent Laplacian, zero eigenvalues only (β_0)	85.8 ± 0.1	50.3 ± 1.6	**Good** — very stable
5	[3] pTU	83.0 ± 5.2	39.4 ± 7.0	**Good but unstable** — misses Gaussian noise
6	[4] Persistent Laplacian, full spectrum	82.6 ± 0.8	59.3 ± 1.5	**OK** — extra eigenvalues make it worse
7	Bayesian mutual information (baseline)	81.4 ± 4.8	60.2 ± 9.0	**Baseline**
8	[2] TU (paper version)	79.1 ± 0.8	73.1 ± 3.1	**OK** — beats confidence
9	Softmax confidence, MSP (baseline)	74.1 ± 4.8	74.1 ± 3.0	**Weak baseline** — fooled by confident mistakes
10	[1] Neural Persistence	55.4 ± 5.0	89.2 ± 4.9	**Fails** — near random guessing

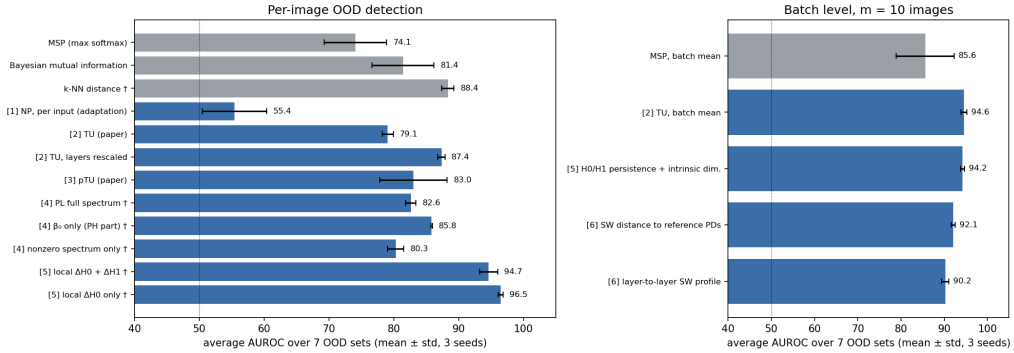


Figure 1. Average AUROC (mean ± std over 3 models). Grey = baselines, blue = topological approaches; † = evaluated on 1000 images per set. Dotted line = random guessing.

4b. Groups of images — is a whole batch of m images from a different source? (average AUROC, 3 models)

Method	10 images	20 images	Comment
[2] TU, averaged over the batch	**94.6 ± 0.7**	96.8 ± 0.5	Best with very small batches
[5] Cloud topology + intrinsic dimension	94.2 ± 0.5	98.9 ± 0.0	Ties TU at 10 images, ahead from 20
[6] Distance between persistence diagrams	92.1 ± 0.4	**99.0 ± 0.2**	Near perfect from 20 images
[6] Layer-to-layer distance profile	90.2 ± 0.9	94.6 ± 1.6	Paper's main tool; weaker for OOD
Softmax confidence, averaged (baseline)	85.6 ± 6.7	89.0 ± 8.3	Lowest and least stable

[3] pTU statistical test (the paper's own test on groups of images): with 50 images it flags every OOD set in 96–100 % of trials, while flagging normal MNIST only 3–5 % of the time (the intended 5 % false-alarm rate). All group methods reach about 100 AUROC with 50 images.

5. Main takeaways

1. **Topology catches what confidence misses.** The network is 94–100 % confident on SVHN, CIFAR-10 and uniform noise, so the confidence score fails there (AUROC 60–69). TU, pTU and β_0 still detect them (78–100), and their results barely change between trained models.
2. **A method must look at the input.** Neural Persistence [1] describes the trained weights, which are the same for every input, so it cannot tell normal and unusual inputs apart.
3. **Simple summaries beat complicated ones.** Counting connected pieces (β_0) works better than the full persistent-Laplacian spectrum and TU's distance; the extra eigenvalues and loop (H1) features add noise.
4. **How layers are combined matters.** The first layer is the most informative in [2], [3] and [4], but those papers average all layers equally. Rescaling each layer before averaging lifts TU from 79 to 87.
5. **The best single-image score is mostly a distance.** [5]'s top result measures how far an image is from its nearest training images; its loop-based part alone is worse than guessing.
6. **For groups of images, everything works.** All group methods reach about 100 AUROC with 50 images; pTU's test is the only one with a controlled false-alarm rate.

6. Limitations and next steps

- **Scope:** one small MLP on MNIST (the setting of [1]–[3]; [5] and [6] studied CNNs); 3 trained models.
- **My adaptations** (labelled above): single-image versions of [1] and [5], the MLP version of [6], rescaled TU, and β_0-only; plus the 0.1 KL scaling.
- **Next:** repeat on a CNN (CIFAR-10), test layer-weighted TU/pTU, and use a two-sided pTU score so low-variance inputs such as Gaussian noise are caught.

References: [1] Rieck et al., ICLR 2019. [2] Lacombe et al., IJCAI 2021. [3] Yeh & Yang, arXiv:2511.18813. [4] Wolf et al., arXiv:2509.20220; Jones & Wei (PETLS); Botnan & Dong, arXiv:2408.16741. [5] Magai & Ayzenberg, arXiv:2204.08624. [6] Purvine et al., AAAI 2023.